

BIOS232 – Assign II (Part 1 of 2): Short Critical Review

Link to the Paper: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10501982/>

Patil, S.S., Ramteke, M., Verma, M., Seth, S., Bhargava, R., Mittal, S. and Rathore, A.S., 2023.

A domain-shift invariant CNN framework for cardiac MRI segmentation across unseen domains.

Journal of Digital Imaging, 36(5), pp.2148-2163.

This work serves as a critique of Patil et al.'s 2023 paper, “*A domain-shift invariant CNN framework for cardiac MRI segmentation across unseen domains.*” Magnetic Resonance Imaging (MRI) is a vital bioimaging modality in cardiology, with most image analysis pipelines requiring manual annotation of anatomical structures. This ‘segmentation’ of images is a time-consuming task requiring expert human operators. Thus, there have been many efforts focused on automating MRI segmentation using Deep Learning (DL) techniques. However, given the tendency of DL models to ‘overfit’ to datasets, this can lead to biases towards the ‘domains’ on which they are trained. For example, if a model is trained and performs well on data from a population ‘A’, it may not perform as well for scans from a population ‘B’. This artefact can lead to reduced performance in clinical deployments and therefore represents an important problem to address. *Patil et al. do so by proposing a DL-based Convolutional Neural Network (CNN) framework that leverages diverse training data to improve domain generalisability in cardiac MRI segmentation, as well as benchmarking the segmentation performance of state-of-the-art (SoTA) architectures. This critique argues that while the study is well-motivated and methodologically sound in its experimental design, it remains limited in its reproducibility and comparative scope.*

This study's primary objective was to design a DL framework that improves the generalisability of CNNs used for MRI segmentation of cardiac structures. The proposed schema leveraged heterogeneity in training datasets to do so. The paper was built upon the premise that, due to so-called 'domain shifts' arising from differences in imaging sequences and patient populations, issues of reliability arise that preclude the widespread deployment of DL architectures such as CNNs in clinical practice.

Three SoTA CNN segmentation architectures—U-Net, Attention U-Net, and Attention-Res-U-Net—were trained on multiple datasets. The ACDC and MSCMR 2019 datasets introduced variance in terms of the sequences derived (including imaging sequences such as bSSFP and T2-weighted) and pathological conditions. Separately, the architectures were trained on a multi-vendor, multi-centre, multi-disease (M&M) dataset to enable comparative analysis of the segmentations produced between multi-sequence heterogeneity and vendor-based diversity. The authors also conducted a comparison between training on the multi-sequence set (ACDC+MSCMR) and training on M&M alone to assess the effect of sequence diversity on cross-domain performance. To reduce reliance on superficial image properties (such as texture, contrast, and intensity), the models were further exposed to heterogeneous imaging characteristics through synthetic augmentation of training data (i.e., intensity shifts, affine transformations, etc.).

The authors evaluated the robustness of each of the three architectures on independent test datasets not included in training (e.g., MYOPS and AIIMS). Segmentation performance was primarily assessed using the Dice coefficient, which measures the degree of spatial overlap between predicted and reference segmentations. The Dice scores show that U-Net was the most consistently “translatable” across unseen datasets, with Attention-U-Net occasionally performing

marginally better on specific structures but not reliably so. The authors attributed the reduced performance of attention-based models to overfitting on image features. The study demonstrated that training on a multi-sequence dataset (ACDC+MSCMR) yielded a 3–4% absolute Dice improvement on the external AIIMS cohort compared to M&M-only training. The authors therefore suggest that sequence heterogeneity contributed more to robustness than vendor diversity alone. The study further derived clinically relevant functional parameters for the Indian cohort from the predicted masks. These included end-diastolic volume (EDV), end-systolic volume (ESV), and ejection fraction (EF). These derived values were shown to statistically agree with the corresponding reference measurements, with high correlations between them ($r > 0.95$ for all three metrics).

From a critical perspective, this work is technically sound and focused on translatability. For instance, the data augmentation strategy was logically aligned with the goal of developing architectures capable of handling variability across imaging conditions. Another methodological strength concerns the authors' attempt to move the work closer to translational relevance. Despite being a standard imaging metric, Dice scores alone do not guarantee clinical usability. The authors therefore went beyond reporting Dice scores by deriving clinically relevant functional parameters from the predicted segmentations and demonstrating statistical agreement with the reference measurements. Another strong aspect of this work was its presentation of a clear and well-articulated case for the motivation. Domain variance in cardiac MRI exists in the form of different imaging sequences, scanner vendors, acquisition protocols, etc. Thus, overfitting is a barrier to automated segmentation and a niche worth exploring. The study addresses fundamental issues with DL systems (such as overfitting) which can actualise as bias,

especially given that radiology literature is usually focused on datasets from the Western world. By showing that their framework generalised well to an independent Indian MRI cohort not included during training, the study helps address a crucial gap in the literature concerning the domain robustness of cardiac MRI segmentation models with respect to population biases.

A key limitation of this study concerns transparency. As is standard practice in computational research, the analytical code should be made publicly available to enable verification, replication, and extension of the findings. Given that the study builds on openly accessible datasets, there appears to be no substantive barrier to sharing the code or models, and its absence raises questions about rigour and weakens its contribution to the scientific community. Moreover, well-established anonymisation and de-identification practices would allow clinical data to be shared in a privacy-preserving manner. Another limitation concerns a narrow methodological comparison. Although it is understood that the study focuses on DL segmentation frameworks, there is no mention or evaluation of alternative paradigms such as atlas/template-deformation (image-registration) approaches. This omission is particularly striking because atlas methods remain important in cardiac MRI segmentation and, in some settings, continue to represent strong or even SoTA comparators. Therefore, a discussion or review of such methods could have provided a useful baseline for contextualising performance.

This study represents a meaningful effort towards more generalisable DL-based cardiac MRI segmentation. It offers practical value in terms of the training strategy, external cohort evaluation, and its extension to clinically relevant measures. The paper seeks to address the problem of domain shift in cardiac MRI segmentation by leveraging heterogeneity in training data to improve cross-domain generalisability. *As argued throughout this critique, the study is*

well-motivated and methodologically well-built; despite these strengths, limitations in reproducibility temper the scope of its contribution, and the work could have benefited from broader methodological comparison.